CAPABILITY OVERVIEW

Madtronix Systems Private Limited

High-Speed Hardware Engineering, Embedded Firmware & Turnkey Prototyping

Headquarters: Noida, India

madtronixsystems.com

CORPORATE PROFILE

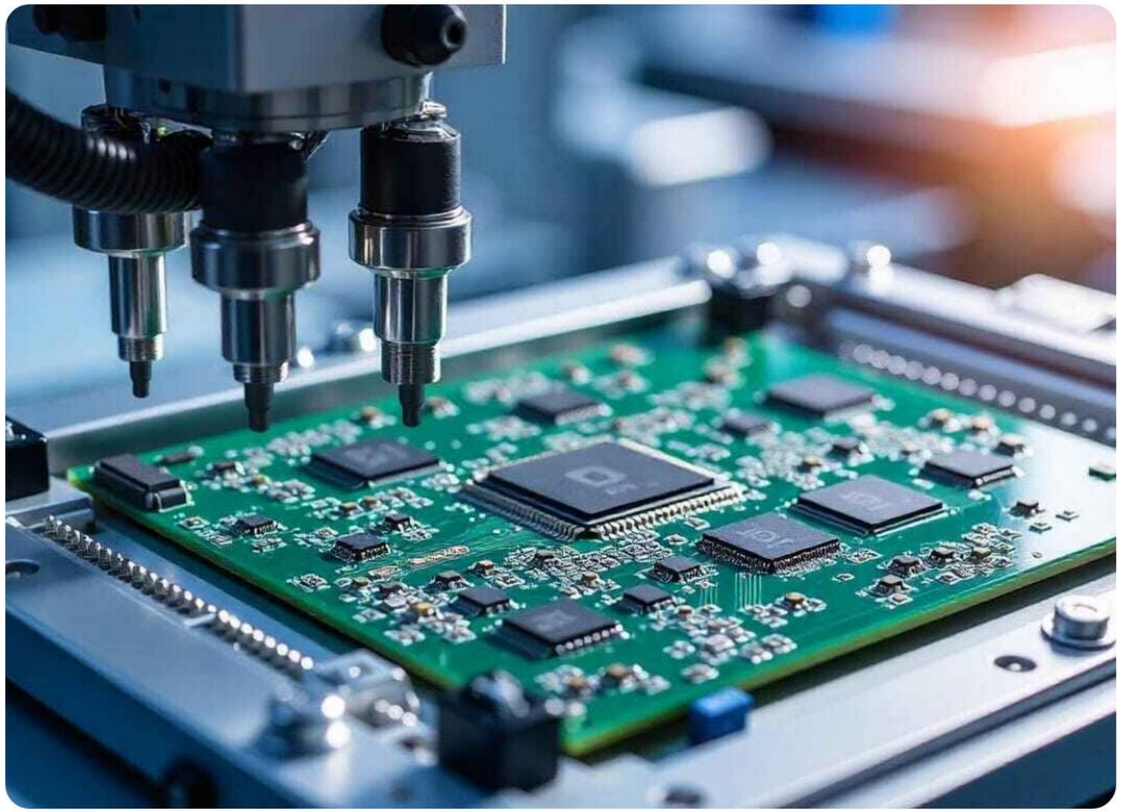
Engineering Core Capabilities

Madtronix Systems delivers comprehensive electronic design services, transforming complex system specifications into production-grade hardware architectures.

Multilayer PCB: High-speed signal routing with controlled impedance.

Firmware Engineering: Bare-metal drivers and RTOS stack integration.

Rapid Prototyping: Quick-turn board bring-up and verification.



SERVICE OFFERINGS

Specialized Engineering Disciplines

Hardware Architecture

Precision schematic capture and BOM optimization.

High-Speed PCB Design

EMI/EMC mitigation and impedance matching.

Embedded Firmware

Low-latency firmware and secure telemetry stacks.

WORKFLOW

End-to-End Development Lifecycle

01

Specification

Architecture freeze & power budget design.

02

Schematic & Layout

Multi-layer routing with signal integrity checks.

03

Board Bring-up

SMD assembly, voltage checks & bare-metal test.

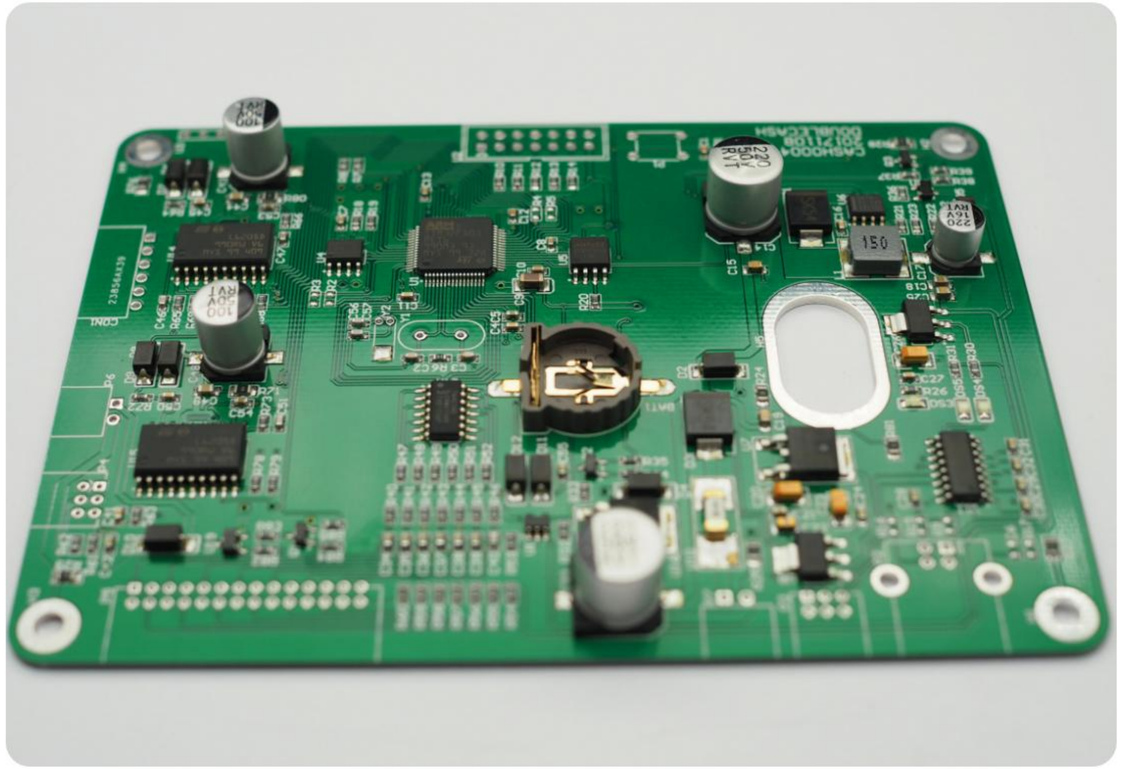
04

Validation

Environmental, thermal & compliance testing.

PROTOTYPING STANDARDS

Hardware Quality and SMT Precision



IPC Class 2 & 3 Compliance

Precision surface-mount soldering with automated optical inspection.

Component Obsolescence Defense

Proactive sourcing and pin-compatible alternate parts analysis.

BOM Cost Engineering

Design optimization targeting high-yield mass production.

TESTING & VERIFICATION

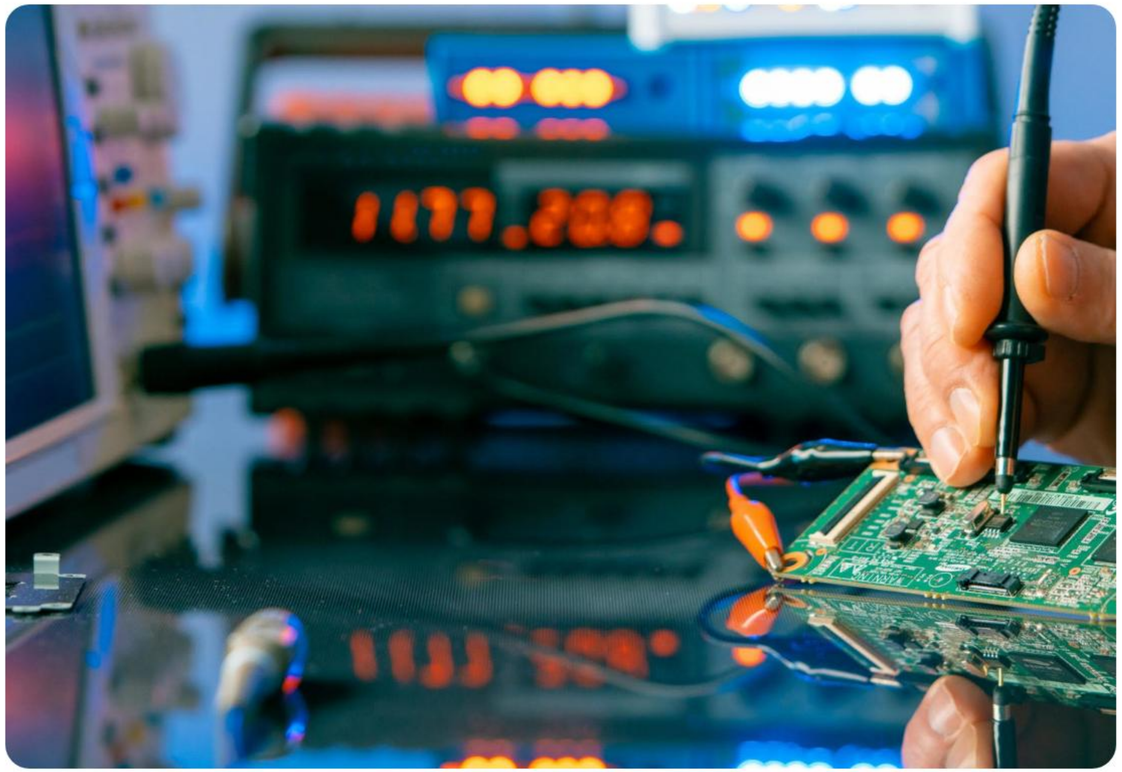
In-House Laboratory Capabilities

Our testing protocols catch signal flaws and thermal stress well before pilot manufacturing commits.

Signal Integrity: High-bandwidth mixed-domain oscilloscopes.

Power Profiling: Microamp-level telemetry power audits.

Thermal Audits: Continuous infrared load monitoring.



MISSION-CRITICAL ELECTRONICS

Defense & Aerospace Engineering

MIL-STD Ruggedization

MIL-STD-810 environmental and shock survivability design.

Tactical Avionics

High-density flight telemetry and servo actuation systems.

Extreme Thermal Limits

-40°C to +125°C conduction cooling architectures.

ADVANCED SPECIALIZATIONS

EMI/EMC, FPGA & RF Engineering

EMI / EMC Certification

MIL-STD-461 & CISPR 32 compliant pre-compliance design.

High-Speed FPGA Design

VHDL/Verilog DSP pipelines on AMD Xilinx & Intel SoC.

RF & Microwave Routing

Sub-6 GHz, Rogers substrates, LNA & PA circuit tuning.

SECTOR APPLICATIONS

Industries Served

Defense & Tactical

Rugged communications, SDR & radar signal units.

E-Mobility & EV

BMS controllers, DC fast chargers & telematics units.

UAV & Drones

Lightweight flight controllers & power distribution.

Industrial IoT

Cellular & LoRa edge gateways for telemetry systems.

ENGAGEMENT MODELS

Flexible Commercial Collaboration

Turnkey Product NPI

Complete ownership from architecture concept to factory-ready Gerber files.

Ideal for OEM product rollouts

On-Demand R&D Burst

Supplemental engineering support for high-density layout and firmware milestones.

Ideal for overloaded hardware teams

Design Optimization

Board revamps for cost reduction, obsolescence fixes, and EMC pass guarantee.

Ideal for active volume products

Image Sources



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